Radioactive Decay Counts: Physics and Exercise Guide for 4B

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Mathematical Methods for Physicists II

Python Notebook Guide

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This is a collection of learning notes accompanying notebook 4B, a radioactive-counting and parameter-estimation extension to my Mathematical Methods for Physicists II computational materials. It introduces the new decay physics, count model, likelihood, and uncertainty concepts before walking through the synthetic experiment. Basic probability and Monte Carlo ideas are referenced from the earlier guides.

1 Where this project fits

Notebook 4B is an extension about radioactive counting and parameter estimation. Its physical and statistical concepts are not supplied by 4A's discrete-map theory, so they are introduced here before the exercise walkthrough. The earlier probability and Monte Carlo notes provide expectations, variance, independent sampling, and repeated experiments; those basics are referenced rather than derived in full.

The central question is how a smooth underlying mean decay rate produces fluctuating integer observations, and how to infer the rate's decay constant from one finite record. This is genuine sampling randomness in the model, not the deterministic irregularity studied in 4A.

All data are synthetic. The background is assumed known and constant, detection is idealized, and the model has no dead time. The notebook demonstrates an inference method, not a measured half-life of an identified isotope.

2 Decay probability and the exponential law

2.1 Constant hazard for one nucleus

Suppose a surviving unstable nucleus has a constant probability per unit time λ of decaying. For a short interval dt, its conditional decay probability is $\lambda\,dt$ to first order. If $S(t)$ is its survival probability,

$$\frac{dS}{dt} = -\lambda S, \qquad S(t) = e^{-\lambda t}.$$

The lifetime density is $p(t) = \lambda e^{-\lambda t}$ for $t \geq 0$. This exponential lifetime is memoryless: survival to an earlier time does not change the subsequent conditional lifetime law. The mean lifetime is $1/\lambda$.

For an initial collection of N_0 independent nuclei, the *mean* surviving number is $\overline{N}(t) = N_0 e^{-\lambda t}$. A finite actual population changes by integer jumps, so a smooth $N(t)$ is an expectation or macroscopic approximation, not each microscopic realization.

2.2 Activity, detected rate, and half-life

Activity is the expected decay rate $\lambda\overline{N}(t)$. A constant detector efficiency η can be absorbed into the observed initial signal rate $R_0 = \eta\lambda N_0$, giving

$$R(t) = R_0 e^{-\lambda t}.$$

The half-life solves $e^{-\lambda t_{1/2}} = 1/2$:

$$t_{1/2} = \frac{\ln 2}{\lambda}.$$

It differs from mean lifetime. The decay constant has inverse-time units, while R_0 and background rate b have counts per unit time. The example uses an unspecified consistent time unit; its printed half-life is in that same unit, not automatically seconds or years.

The population tends to decay on average, while measured *bin counts* need not decrease at every step. A late count can exceed an earlier count without a nucleus regenerating or the physical law failing.

3 Poisson counting and its limits

For independent ideal events with a prescribed rate, the count in an interval is modeled by

$$P(K = k \mid \mu) = \frac{e^{-\mu}\mu^k}{k!}, \qquad k = 0, 1, 2, \ldots,$$

where μ is the expected count integrated over that interval. Its mean and variance are both μ, so its standard deviation is $\sqrt{\mu}$. Its fractional standard deviation is $1/\sqrt{\mu}$ for $\mu > 0$.

The rate need not be constant within a bin: a nonhomogeneous Poisson process permits a specified varying intensity and still gives Poisson counts with mean equal to its integral. Counts in disjoint bins are independent

under that model. Independent Poisson signal and background contributions add to another Poisson count with the sum of their means.

These are assumptions, not consequences of writing an exponential curve. For a fixed finite collection of nuclei each nucleus can decay only once. Counts allocated to disjoint intervals then follow a multinomial construction, not automatically independent Poisson counts. If p_i is one nucleus's probability of decay in bin i, their covariance is $-N_0 p_i p_j$ for different bins. Rare observed events from a large source can justify a Poisson approximation; a Poisson-distributed initial population offers another route to independent Poisson bin counts.

The notebook explicitly adopts the independent-Poisson observation model. Detector dead time, varying efficiency, correlated backgrounds, or significant finite-source dependence would require changing it for real data. The model's limitations are different from numerical-integration errors in 2A or 2C.

4 Integrate rates before comparing with counts

The ith bin extends from t_i to $t_i + w$, where $w = \Delta t$. Its expected count is

$$\begin{aligned}\mu_i &= \int_{t_i}^{t_i+w} [R_0 e^{-\lambda t} + b], dt \\ &= \frac{R_0}{\lambda} e^{-\lambda t_i}(1 - e^{-\lambda w}) + bw.\end{aligned}$$

This has count units. Evaluating $R(t)$ at a bin centre without integrating would compare a rate with a count and give incorrect normalization. Multiplying the centre rate by width can be an approximation for narrow bins, but the exact expression is simple here.

The implementation uses

$$1 - e^{-\lambda w} = -\operatorname{expm1}(-\lambda w).$$

The `expm1` function accurately computes $e^z - 1$ for small z without subtracting nearly equal floating-point numbers. Thus it reduces cancellation when λw is small.

In the limit $\lambda \to 0$, the signal mean approaches $R_0 w$. The current function divides by λ and is used only with positive values enforced by the fit bounds; it does not implement an exact zero-decay branch. This is consistent with the stated experiment, not an invitation to call it at zero unchanged.

Each bin can be plotted at its centre $t_i + w/2$ for readability, while the model still integrates from the left edge. For contiguous bins, the sum of expected counts equals the integral over their combined observation period, a useful independent check.

5 Likelihood: infer parameters from the observation model

5.1 From probabilities to an objective

For observed counts k_i, independent bins give likelihood

$$L(R_0, \lambda) = \prod_i \frac{e^{-\mu_i} \mu_i^{k_i}}{k_i!}.$$

Here the data are fixed and the parameters vary through μ_i. Taking logarithms converts products into sums. Maximizing likelihood is equivalent to minimizing

$$\mathcal{L}(R_0, \lambda) = \sum_i [\mu_i - k_i \ln \mu_i],$$

after dropping the parameter-independent $\sum_i \ln(k_i!)$. This omission preserves the fitted maximizer but does not make the resulting objective an absolute normalized probability or a standalone goodness-of-fit statistic.

For a zero-count bin, the contribution is simply μ_i. Its probability $e^{-\mu_i}$ still constrains overly large model means. Zero observations are therefore informative and should not be discarded or replaced by ones.

5.2 Why a straight-line log fit is inappropriate here

With known background the *expected* signal rate has a logarithm linear in time. Observed counts, however, fluctuate, may be zero, and can be smaller than the expected background count. Taking $\ln(k_i - bw)$ can be undefined and transforms the noise nonlinearly. Fitting raw nonnegative counts with the correct Poisson likelihood uses all bins without this problem.

The unknown amplitude and decay constant are fitted jointly, so either can trade against the other within the data constraints. Fixing the amplitude to its synthetic truth would be an easier but different inference problem. The true values are benchmarks for evaluation, not inputs to the likelihood fit.

5.3 Bounded optimization and checks

The notebook uses SciPy's L-BFGS-B optimizer, which handles parameter bounds. The search limits are $0.001 \leq R_0 \leq 500$ and $0.0001 \leq \lambda \leq 2$. These maintain positive model means with the chosen positive background; they are computational search restrictions, not uncertainty intervals or a Bayesian prior distribution.

Convergence status must be checked. If a fit reaches a boundary, fails, or depends strongly on the starting guess, investigate before treating it as an ordinary interior maximum. More starting points can test numerical robustness, but a successful local search alone does not establish correctness of the observation model.

6 Count spread, parameter uncertainty, and residuals

6.1 Three different questions

The Poisson standard deviation $\sqrt{\widehat{\mu}_i}$ describes counting spread at a fitted mean. An uncertainty band for the *mean curve* would propagate uncertainty in the fitted parameters. A prediction interval for a new count would also account for the discrete distribution and, when appropriate, parameter uncertainty. These are not interchangeable.

The plotted mean-plus-or-minus-square-root band is only a counting-spread reference. Clipping its lower edge at zero acknowledges the count domain but does not turn it into an exact probability interval. At small means, a Poisson distribution is asymmetric and discrete, so Gaussian one-sigma coverage intuition is unreliable.

6.2 Pearson-style residuals

The standardized residual is

$$z_i = \frac{k_i - \widehat{\mu}_i}{\sqrt{\widehat{\mu}_i}}.$$

If the *true known* means were used under the ideal model, each residual would have mean zero and variance one, even though its distribution need not be Gaussian. With fitted means, the parameters depend on all the data, changing these properties and inducing relationships among residuals.

Look for systematic time patterns, not exact point-by-point agreement with a Gaussian sketch. A residual mean near zero and RMS on the scale of one are rough checks. They are not a calibrated hypothesis test, and fitted residual RMS need not equal one. A proper goodness-of-fit assessment could simulate and refit repeated datasets and compare a chosen diagnostic; this notebook does not calibrate such a test.

Use fitted expected counts in the denominator, not square roots of observed counts. The latter would divide by zero for empty bins and give data-dependent weighting with a different meaning.

7 Parametric bootstrap: repeat the fitted experiment

7.1 What is being simulated

After obtaining $(\widehat{R}_0, \widehat{\lambda})$, compute fitted means $\widehat{\mu}_i$. For each bootstrap replicate draw new independent counts $K_i^* \sim \text{Poisson}(\widehat{\mu}_i)$ on the *same* time grid, then refit both unknown parameters. The collection of $\widehat{\lambda}^*$ values approximates the estimator's sampling variability under the fitted model.

This is parametric because new data come from an assumed probability model. It is not nonparametric resampling of existing bins: those bins have different expected counts and should not be treated as exchangeable observations of one fixed mean. It is also not posterior sampling of the unknown parameter.

Keeping background fixed means the spread is conditional on its assumed known value. A real separately measured background has uncertainty, which would need its own joint model or propagation. Bootstrap repetition does not correct a wrong decay model, an unmodeled detector effect, or a biased fit procedure.

7.2 Percentiles and the half-life transformation

The notebook uses 100 replicates and reports the 2.5th and 97.5th percentiles as a rough percentile-bootstrap interval. One hundred samples leave very few observations in each tail, so these endpoints have noticeable simulation noise. More replicates would stabilize the tail estimates but would not repair model misspecification.

The interval is not a statement of 95% posterior probability for this dataset's true parameter. Frequentist coverage concerns repeated datasets, and simple bootstrap percentile intervals need not achieve nominal coverage in small, biased, or boundary-affected problems.

Because $t_{1/2} = \ln 2/\lambda$ decreases with λ, a positive decay-constant interval $[\lambda_{\text{lo}}, \lambda_{\text{hi}}]$ maps to

$$\left[\frac{\ln 2}{\lambda_{\text{hi}}}, \frac{\ln 2}{\lambda_{\text{lo}}}\right].$$

Reversing the endpoint order is required. The resulting half-life interval is generally asymmetric even when a decay-constant interval looks nearly symmetric.

8 Walkthrough 1: test fixed-rate Poisson counts

Use expected count nine and generate 2,000 independent bins. Evaluate the Poisson probability at integer k values from zero through 22, using the exponential, integer power, and factorial. Build width-one histogram bins centred on the integers.

With unit-width bins, density-normalized histogram heights equal empirical probabilities if the entire observed sample is included. The current sample has maximum count 20, so the displayed support contains it. The unplotted theoretical tail above 22 remains nonzero; a different experiment with larger observed counts should extend the histogram rather than silently condition on a truncated sample.

The reproduced run gives mean about 8.940, sample variance about 8.923, and sample standard deviation about 2.987, compared with predictions nine, nine, and three. Finite-sample fluctuations explain the lack of exact equality. These stationary bins are a preliminary probability check, not the subsequent declining-rate experiment.

9 Walkthrough 2: generate the decay record

Use width $w = 0.5$ and left edges from zero to 19.5, making 40 contiguous bins ending at time 20. The inputs are $R_0 = 60$, $\lambda = 0.22$, and known background $b = 2$. The background contributes one expected event per bin. The generating half-life is $\ln 2/0.22 \approx 3.151$ time units.

Calculate one integrated mean per bin using the common function, then draw one count for each mean. Plot observations and means at the bin centres, including the horizontal background-count reference. The example produces 318 total observed events and two zero-count bins.

The input-mean curve in this first plot uses information known because the experiment was simulated. It is not yet a fitted model inferred from the observations. Apparent late upward fluctuations are compatible with the decaying mean, especially as signal counts approach the background level.

10 Walkthrough 3: fit both unknown parameters

Construct a negative-log-likelihood function taking the parameter pair, observations, left edges, width, and known background. Within it call the same expected-count function used for simulation. This prevents a mismatch between integrating data generation and fitting an approximate point-rate model.

The initial amplitude guess is $\max(k_0/w - b, 1)$. It approximates a rate from the first bin and keeps the starting value positive; it is not itself the inferred amplitude. The initial decay guess is 0.2. Minimize the objective with the stated bounds, check success, and unpack the fitted pair.

The reproduced values are $\widehat{R}_0 \approx 56.001$, $\widehat{\lambda} \approx 0.1965$, and fitted half-life about 3.527. The differences from 60 and 0.22 are plausible sampling variation in one realization; uncertainty must be assessed separately. The fitted values are interior to the search bounds, so these particular results are not boundary solutions.

Plot the integrated fitted means with the same observations. A fitted curve should summarize the mean trend, not pass through every point. The true parameters have not been substituted into this fitting step.

11 Walkthrough 4: inspect counting fluctuations

Take square roots of the fitted bin means and form the standardized residuals. Plot the counting-spread reference band alongside the observed counts, and separately plot residual against time with zero and ± 1 reference lines.

The reproduced residual mean is about -0.013 and RMS about 0.667. Do not rescale the counting model merely to force the RMS to one, or declare model rejection from this value alone. Counts are low in late bins and two parameters were estimated from the same 40 observations; the diagnostic is not exactly an independent standard-normal sequence.

An evident time trend could motivate revisiting the background, efficiency, or decay model. No trend would still not prove every assumption. Distinguish such model checks from uncertainty in λ, addressed by the next experiment.

12 Walkthrough 5: bootstrap and transform the interval

Initialize an empty decay-estimate list. Repeat 100 times: draw counts from the fitted means, fit both amplitude and decay constant using the same procedure, and save the new decay estimate. Reusing the same seeded stream without resetting it generates fresh experiments reproducibly.

Plot a histogram of the 100 estimates with vertical markers for the original fit and the synthetic generating constant. Compute the two percentiles and transform the half-life endpoints in reverse order.

For the current run the decay-constant range is approximately 0.1648 to 0.2280, corresponding to half-life about 3.040 to 4.207. These exact rounded values depend on the random stream, including earlier draws. They illustrate uncertainty conditional on the assumed model; they are not universal uncertainty bars for every dataset with this duration.

Every bootstrap fit converges in the reviewed run. In another parameter regime, failed or boundary fits should be diagnosed rather than silently discarded, because selectively retaining only convenient fits can bias the resulting histogram.

13 Checks and the scope of the extension

Stage	Check	Interpretation limit
Decay model	Positive rate and integrated count units	Finite-source independence is assumed, not automatic
Simulation	Forty bins and fixed known background	Synthetic parameters are not experimental measurements
Fit	Positive means, convergence, bounds, same bin model	Optimization success is not goodness of fit
Residuals	Expected Poisson variance and no obvious time pattern	Low-count residuals need not be Gaussian
Bootstrap	Fresh counts and refit both unknowns	Conditional model-based interval, not a posterior

No change to the decay-count model or fitted calculations was needed in this review. The integrated-count equation's notation was corrected, important implementation steps were commented, and explanatory discussion was added after the probability, fit, residual, and bootstrap plots.

For real measurements, explicitly include background uncertainty, efficiency changes, and dead time when relevant. The new physics in this extension is stochastic decay counting and inference, not deterministic chaos from the preceding map notebook.